

# Speaking Script + Q&A Bench — 19-slide grayscale talk

**Deck:** CodaWork2026\_Presentation\_2026-05-27.pptx — single grayscale file, **19 slides**, every slide numbered N / 19.

**Narrative (clinical order):** standard CoDa → how we add **time** to the simplex (the five named readings) → three highlighted countries → **the rest of the world** (the other six) → the live instrument closes the show.

**Timing:** ~13 min spoken across 19 slides (the rest-of-world finale is a brisk sweep) · +~1 min live HTML close = **~14 min** · then **5 min Q&A**.

**How to read this page:** left column is what you say (read from this). Right column is the Q&A bench — likely questions with ready responses. Glance right only if a question lands.

**Voice notes:** No preamble. Periods, not commas. Numbers first, qualifier after. Slow on the three case studies (slides 6–11). The rest-of-world finale (13–18) is a sweep — one breath per country, do not dwell. End on slide 19 and open the projector.

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## Slide 1 · Title — standard CoDa, and we add time · 40 sec

| Speech (read this)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Q&A bench                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <p>Good morning. <i>Compositional monitoring of energy-mix drift on the simplex.</i></p> <p>The question: <b>which carrier did the structural work?</b> Not which one got bigger — which one moved the composition.</p> <p>This is standard compositional data analysis. The one new move is what happens when we add time to the simplex.</p> <p>Follow along on the repository — the deck, the manuscript, and the live projector are all open. Every view here is reproducible on your data.</p> | <p><b>Q:</b> Who are you / what's BTL? <b>A:</b> Audio/electronics engineer; founder of the Binaural Test Lab at Rogue Wave Audio, Markham, Ontario. H<sup>s</sup> generalised retroactively from acoustic compositional work.</p> <p><b>Q:</b> How long is the talk? <b>A:</b> ~13 minutes across 19 slides, then ~1 minute on the live projector — about 14 minutes — then 5 minutes of Q&amp;A.</p> <p><b>Q:</b> "Any compositional dataset" — what's included? <b>A:</b> Anything CoDa-describable: closed, non-negative, finite carriers, static or time-series. Energy mixes, biogeochemistry, microbiome ratios, expenditure shares, CMB photon power per multipole. Three IEEE-floor reference datasets in the repo confirm engine behaviour across unrelated domains.</p> <p><b>Q:</b> Why is the deck grayscale? <b>A:</b> High-contrast at a distance and low-cost to reproduce. The structure carries the meaning; colour was doing no work the labels don't already do.</p> |

## Slide 2 · The size view hides the work · 70 sec

| Speech (read this)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Q&A bench                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <p>Twenty-five years of German electricity. Stacked-area chart. Coal and lignite recede. Nuclear phases out. Gas holds. Solar and wind grow steadily. Wind passes coal in the late twenty-tens.</p> <p>That is the headline view. Here is what it hides.</p> <p><b>Germany Solar, 2005 → 2006.</b> Starting share, zero-point-two-one percent. Structural Power Share, seventy-one-point-one percent. Activation Coefficient, approximately three-hundred-thirty-three times.</p> <p>Solar acted at three-hundred-thirty-three times its size — four years before the share view calls solar visible. No size view shows that.</p> <p>This talk is the reason that number exists. The mathematics is standard compositional data analysis. The application — monitoring the simplex for structural work — may be new.</p> | <p><b>Q:</b> What dataset? <b>A:</b> EMBER electricity, public CC BY 4.0. Country-level annual generation by carrier 2000–2025. Hash-chained to the source CSV; reproducible.</p> <p><b>Q:</b> Why Germany Solar 2005–2006? <b>A:</b> First step where Solar’s Power Share crosses 70 % in the German grid — the first year structural work concentrates in solar, four years before its size-view share crosses ~1 %.</p> <p><b>Q:</b> Isn’t 333× just a tiny denominator? <b>A:</b> Yes, by construction. That sensitivity <i>is</i> the diagnostic — the work-to-size ratio. The framework names when a small carrier is structurally large.</p> <p><b>Q:</b> Could this be noise? <b>A:</b> No — 0.21 % is recorded share, not estimate. CLR motion is mathematically defined and reproducible byte-identically.</p> |

## Slide 3 · Reading time on the simplex — the five named methods · 60 sec

| Speech (read this)                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Q&A bench                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <p>Here is the whole method. Standard CoDa geometry — but we walk it forward, year by year. Five named readings.</p> <p><b>Composition</b> — what share each carrier has. <b>Helmsman</b> — which carrier makes the largest CLR move at a step. <b>Helmsman trajectory</b> — when the largest-motion carrier changes. <b>Power Share</b> — how much squared CLR motion each carrier did. <b>Activation Coefficient</b> — Power Share divided by starting share.</p> | <p><b>Q:</b> Why exactly five? <b>A:</b> Each answers a distinct question the others don’t. Drop one and you lose a question. Drop all five and you have a stacked-area chart.</p> <p><b>Q:</b> How is this different from PCA / SVD? <b>A:</b> Not dimensionality reduction. Each reading is interpretable per-carrier per-step. Diagnostic, not compressive.</p> <p><b>Q:</b> Why ILR-Helmert? <b>A:</b> Orthonormal basis with no preferred carrier; rotation-invariant.</p> |

All five derive from CLR plus Helmert-ILR. Standard geometry — no new mathematics. The new move is reading it over time. That is the whole talk; the rest is showing it work.

Standard CoDa basis since Egozcue et al. 2003.

**Q:** Is this in the manuscript? **A:** Yes — Appendix A equations 1–8 + Supplementary §S2 has the full formula tables.

## Slide 4 • The Activation Coefficient • 60 sec

### Speech (read this)

Alpha-sub- $i$  of  $t$  equals Power Share over starting share. That ratio is the diagnostic.

Alpha near one — the carrier does work proportional to its size. Ordinary. Alpha much greater than one — hidden driver; the carrier acts above its size. Alpha less than one — coasting.

Worked example. Germany Solar, 2005 to 2006. Starting share zero-point-two-one percent — small. Power Share seventy-one-point-one percent — most of the work. Alpha approximately three-hundred-thirty-three.

Yeast is two percent of a loaf by mass and does one hundred percent of the rising. Same shape. That is the structural beginning of the Energiewende — named four years before solar appears in the share view.

### Q&A bench

**Q:** Is this just elasticity? **A:** No — elasticity is response to price. Activation Coefficient is structural work over share. Compositional, not behavioural.

**Q:** Doesn't this just exaggerate small carriers? **A:** By design. A small carrier doing small work has  $\alpha \approx 1$ , ordinary; the framework only flags when the work-to-size ratio is large.

**Q:** Threshold for “much greater than 1”? **A:** No hard default. Distribution is heavy-tailed; the framework reports  $\alpha$  directly and lets domain expertise interpret.

**Q:** Does it generalise beyond Germany Solar? **A:** Yes — Japan (multiple high-Activation-Coefficient moments post-Fukushima) and the UK coal exit show the same shape in different regimes.

## Slide 5 • Three archetypes — one instrument, three regimes • 35 sec

### Speech (read this)

Three case studies. Three different transition regimes.

Germany — deliberate transition, a continuous arc. Japan — external shock, loop and reorganise. United Kingdom — regime change, jump and return.

### Q&A bench

**Q:** Why these three countries? **A:** Three textbook cases of three distinct regimes. Representative, not selective — slide 12 gives the corpus-level result on all nine.

**Q:** Did you cherry-pick? **A:** No. The protocol was fixed before dispositions were known; the same five-reading stack applied to all nine countries.

One instrument reads all three. Each country gets a pair of slides next: the share-and-work view, then its trajectory on the simplex.

## Slide 6 · Germany — share and work · 60 sec

### Speech (read this)

The Energiewende. Twenty-five years of deliberate composition change.

The size view tells you what you already know — coal recedes, gas holds, solar and wind grow. The Power Share view tells you something it doesn't.

**Solar 2005 → 2006.** Share 0.21 percent. Structural work 71.1 percent. Alpha approximately 333.

Three years before the size view calls solar visible, the instrument names solar as the structural beginning of the Energiewende. The transition started in the data before it appeared on the chart.

*Pause — flip to the trajectory.*

### Q&A bench

**Q:** Why is Solar 2005–2006 the inflection? **A:** First step where Solar's Power Share crosses 70 % — structural work concentrates in one carrier, three years before its share crosses 1 %.

**Q:** Is this just subsidy timing? **A:** The diagnostic reads composition, not policy. It tells you *when* the composition started changing; *why* is domain knowledge.

**Q:** Why call it deliberate? **A:** The trajectory (next slide) is a smooth monotone arc — no loops, no flips. The geometry of a planned trajectory.

## Slide 7 · Germany — the trajectory · 40 sec

### Speech (read this)

The Germany Helmsman trajectory. Trajectory directness 0.41 — a continuous arc toward the renewable vertex.

Thirteen Helmsman flips across twenty-five years. Smooth, monotone reorientation — no loops, no backtracks. The simplex shows what a deliberate transition looks like: a single sustained trajectory, year after year, in one direction.

### Q&A bench

**Q:** Trajectory directness — what is 0.41? **A:** End-to-end Aitchison distance ÷ total path length. 1.0 = straight line; lower = more meander.  $0.41 \approx$  the composition travelled  $\sim 2.4\times$  the end-to-end distance.

**Q:** Why "Helmsman"? **A:** The carrier with the largest CLR move at a step — the one in largest motion right now.

**Q:** Can I see this live? **A:** Yes — projector at the close. Click DEU → BARY.

This is the geometry of policy intent. Next: what happens when policy doesn't get to choose.

## Slide 8 · Japan — share and work · 60 sec

### Speech (read this)

Fukushima 2011. The shock appears in every reading at once. Composition — nuclear collapses. Helmsman — seventeen flips across twenty-five years, the loudest count in the corpus. Power Share — a massive concentration of squared CLR motion at one step. Aitchison distance 2011 to 2012, about three times the neighbouring-year baseline.

But the years *after* the shock tell the deeper story. Japan's mix kept reorganising for more than a decade — solar, gas, wind, each taking turns absorbing the displaced nuclear share.

The instrument detects both the shock and the multi-year reorganisation that followed.

*Pause — flip to the trajectory.*

### Q&A bench

**Q:** Could the framework predict Fukushima? **A:** No — it detects, it doesn't predict. What it adds is the multi-year *reorganisation* signature that traditional analyses miss.

**Q:** Why 3× baseline distance? **A:** Aitchison distance per step; baseline = median year-to-year step. The 2011→2012 step is an outlier by ~3×.

**Q:** Why isn't nuclear the helmsman every year after 2011? **A:** Helmsman is the carrier *moving most* at a step, not the one in crisis. Once nuclear stabilises low, the largest-motion role passes to whichever renewable absorbs the load that year.

## Slide 9 · Japan — the trajectory · 40 sec

### Speech (read this)

The Japan Helmsman trajectory. Trajectory directness 0.09 — the loop-and-reorganise pattern.

Compare it to Germany's smooth arc. Japan's trajectory revisits and reroutes. The system has to *find* a new composition rather than walk a planned one; the trajectory shows the search.

### Q&A bench

**Q:** 0.09 — what does that mean physically? **A:** The trajectory loops back ~11× the end-to-end displacement. The system explores many states before settling.

**Q:** Is this dynamical-systems language? **A:** Yes — basin-search dynamics rather than limit-cycle or fixed-point. The regime taxonomy is topological-invariant per the Helmsman family.

**Q:** Could noise produce this? **A:** No — see the supplementary null-model. A loop this strong needs correlated perturbation across carriers over years.

That is what an unplanned reorganisation looks like on the simplex. Not a direction — a basin being explored.

## Slide 10 • United Kingdom — share and work • 60 sec

### Speech (read this)

Coal exit as regime change. Between 2012 and 2020, coal goes from more than 30 percent of UK electricity to less than 2 percent.

A true regime change, not a drift. The size view shows the headline cleanly. The Power Share view tells us *how the displaced structural work was absorbed*.

Wind. Solar. Other renewables. Each took portions of the displaced load. There was no single replacement carrier.

The protocol separates size decline from who absorbed the structural work. A regression on raw shares would not show that separation.

*Pause — flip to the trajectory.*

### Q&A bench

**Q:** “Regime change” vs drift? **A:** Aitchison-step magnitude plus the topological signature in the Helmsman trajectory. Coal exits the helmsman role abruptly. Visible on slide 11.

**Q:** Evenly distributed? **A:** No — wind led, but biomass and solar took meaningful shares. The instrument names *who* absorbed work each year.

**Q:** Policy-driven? **A:** UK Climate Change Act + carbon pricing + market dynamics. The instrument reads composition; policy is downstream of what it shows.

## Slide 11 • United Kingdom — the trajectory • 40 sec

### Speech (read this)

The UK Helmsman trajectory. Trajectory directness 0.36 — the jump-and-return pattern.

The trajectory leaves the coal vertex sharply, then settles toward a new mix. Regime change reads as one decisive displacement followed by re-stabilisation. Distinct from Germany’s continuous arc, distinct from Japan’s looping search.

### Q&A bench

**Q:** 0.36 vs Germany 0.41 vs Japan 0.09 — is the UK “in between”? **A:** Quantitatively yes; geometrically distinct. A sharp displacement *then* re-stabilisation — a different topological signature from both.

**Q:** Only three archetypes? **A:** Three illustrative. The framework also recognises drift, oscillation, fixed-point, periodic, and chaotic regimes. Three is what fits a short talk.

Three transitions. Three archetypes. One geometry that names them all.

## Slide 12 · Across the corpus — deceptive drift in 5 of 9 · 55 sec

### Speech (read this)

From three case archetypes to a corpus-level result. We applied the same protocol — same engine, same definition of deceptive drift — to all nine EMBER countries.

Deceptive drift is present in five of nine. **Australia, China, United Kingdom, India, Japan.** It is *absent* in Germany at annual grain, France, the USA, and the World aggregate.

This matters. A useful detector should not flag every system. Discrimination is itself evidence the protocol reads real structure, not artifact.

You have seen three. Here is the rest of the corpus.

### Q&A bench

**Q:** Why these nine? **A:** Pre-selected for grid-level diversity. EMBER public data, hash-chained provenance. Selection fixed before disposition results.

**Q:** Could you have picked to fit? **A:** No — protocol pre-registered, dispositions discovered. 9-of-9 would be indiscriminating; 0-of-9 inert. Five is the result.

**Q:** Why isn't deceptive drift detected in Germany at annual grain? **A:** Its trajectory is mostly monotone (slide 7); annual grain misses moments-within-years. Sub-annual grain might detect it — open question.

## Slides 13-18 · The rest of the world — the other six · ~2 min 15 sec total (one breath each)

*This is the finale: a brisk sweep across the six countries you did not walk through — three with deceptive drift present, three with it absent. Do not dwell. The one-line prompt on each slide is your cue; say it and move.*

### Speech (read this)

**13 · Australia (deceptive drift present).** Same protocol, the rest of the world. Australia — the full trajectory; the deceptive-drift signature, structural work concentrated in the renewable turn.

### Q&A bench (covers all six)

**Q:** Why only the trajectory for these six, not the full plate set? **A:** This is the breadth sweep — the trajectory is the one view that reads at a glance. The full engine plates (all 27 per country, plus Stage 2/3 and CNQ) are in the 325-page master output in the repo.

**Q:** Why does the World aggregate show none when its constituents do? **A:** Large-N smooths fluctuations; sub-population structure averages out. The aggregate composition is more stable than any constituent — itself a finding.

**14 • China (deceptive drift present).** China — coal-era growth, then a turn. The flips pile up; small carriers absorb structural work over and over.

**15 • India (deceptive drift present).** India — the third case where it is present. The same concentration signature, a different grid.

**16 • France (absent).** Now the three with no deceptive drift. France — nuclear-stable across the whole window. The trajectory barely moves; the detector correctly stays silent.

**17 • United States (absent).** The United States — a large grid; no deceptive drift at annual grain. Real moments exist within years; the annual view averages them out.

**18 • World (absent).** And the world aggregate. Large-N smoothing hides the deceptive drift in its constituents — the corpus average is calmer than any country in it.

Nine countries. Five show deceptive drift, four do not. One instrument read them all. *Move to the close.*

**Q:** Australia/China/India — what makes deceptive drift present? **A:** The deceptive-drift composite signature: a structural-work concentration the size view doesn't show, confirmed across the five readings. Per-country detail in the repo.

**Q:** Can you pull up any of these live? **A:** Yes — the projector runs every country in the corpus. Click the country code → BARY at the close or in Q&A.

**Q:** Is the present/absent split robust to grain? **A:** At annual grain, this is the split. Sub-annual grain is an open question flagged in the roadmap — France and the USA are the obvious re-tests.

## Slide 19 • What the stack answers — the live instrument closes • 45 sec + ~1 min live

### Speech (read this)

One observable. Five distinct questions.

**WHAT** carriers are big — the size view. **WHO** is in largest motion — the Helmsman. **WHEN** the largest-motion carrier changes — the Helmsman trajectory. **HOW MUCH** work each carrier did — Power Share. **WHY** a small carrier mattered — the Activation Coefficient.

### Q&A bench

**Q:** Is the projector online / can I get it? **A:** Single self-contained HTML file in the repo (CODA-Association/CODAwork2026/data\_outputs/codawork2026\_projector.html). No server, no build step. Apache-2.0.

**Q:** What are the three modes? **A:** RADAR (per-carrier star plot), BARY (trajectory on the simplex), ALIGN (the Helmsman / largest-motion

The stack does not replace interpretation. It gives interpretation a reproducible object.

And everything you have seen is live. *Open codawork2026\_projector.html*. Same engine output, in the browser — one file, no server. Three modes: RADAR, the composition; BARY, the trajectory; ALIGN, the helmsman. *Click DEU → BARY* — Germany’s arc. *Click JPN → BARY* — Japan’s loop. Same instrument, two regimes, side by side. It runs on any country here, and on your data when you point it at a CoDa series.

Thank you. Questions.

carrier). SHOCK is an overlay that recolours year markers on shock years.

**Q:** Code availability? **A:** Apache-2.0 code + CC BY 4.0 docs. Every algorithm in four forms: Python + R + pseudocode + HUF-STD-002 spec. Three IEEE-floor confirmation datasets verified.

**Q:** How do I run it on my data? **A:** ai-refresh/CCTT\_RUNBOOK.md — 7-phase reproducible runbook. Pseudocode at HCI-CNT/engine/CNT\_PSEUDOCODE.md.

**Q:** Can I collaborate / cite? **A:** Yes. CITATION.cff in the repo. Non-contact discipline: expect either no reply or a substantive one; no follow-up.

General Q&A bench — non-slide-specific questions

| Question                                                          | Ready response                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MC-4 (falsifiability) — what’s the claim, what defeats it?</b> | Three conjuncts: Aitchison-native compositional metrics + formal change-point detection + carrier-level structural-work attribution. Four defeat paths: prior-art, metric, case, category. Any one narrows or kills the claim. Please find one — that’s how the work improves.                                            |
| <b>Why isn’t this already standard CoDa?</b>                      | The mathematics is standard since Aitchison 1986. What I added is the time-series operational layer — walk the geometry forward, ask which carrier did the work at each step, name the diagnostic. That layer is the contribution.                                                                                        |
| <b>Why grayscale?</b>                                             | High contrast at a distance and cheap to reproduce. Structure carries the meaning here; the carrier labels do the work colour would. A scientific-paper aesthetic on purpose.                                                                                                                                             |
| <b>Where does H<sup>s</sup> come from? What’s BTL?</b>            | Binaural Test Lab — acoustics laboratory I founded in Markham, Ontario. The framework generalised retroactively from BTL acoustic work — DADC (2024) → DADI → ADAC (2025) → H <sub>1</sub> operator (2026) → HUF → H <sup>s</sup> . The 6.02 dB cabinet-edge diffraction budget was the first natural simplex constraint. |
| <b>How is reproducibility verified?</b>                           | Trust by independent reproduction. Every algorithm in four forms: Python + R + pseudocode + HUF-STD-002 spec. Re-implement from the pseudocode, verify per-field at IEEE-floor tolerance on the three confirmation datasets. See TRUST_AND_VERIFICATION.md.                                                               |
| <b>AI use — what did you use AI for?</b>                          | HUF AI Collective protocol per HUF-STD-001 v1.1. Claude, ChatGPT, Grok for drafting, review, cross-check, code sweeps. Author byline is human-only; I retain full scientific responsibility. Declaration on slide 19.                                                                                                     |

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|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Is there a manifold underneath this?</b>      | Yes — layered. Smooth underlying object (open simplex with Aitchison structure; CNQ phase space $S^3 \cong \text{SU}(2)$ ); discrete sampling and HTML rendering are piecewise-linear; regime taxonomy is topological-invariant. Working note MANIFOLD_CATEGORY_OF_Hs_PROJECTION.md.                          |
| <b>Is this gauge theory?</b>                     | Substantial gauge-theoretic structure is present in pieces. Closure ( $\Sigma p_i = 1$ ) is a Ward identity; CLR is gauge fixing of the $\mathbb{R}_+$ rescaling symmetry; CNQ's $S^3 \cong \text{SU}(2)$ is the simplest non-abelian gauge group. Consolidated in papers/in_progress/GAUGE_THEORY_AND_Hs.md. |
| <b>If the question is hostile or sweeping...</b> | Don't defend the framework. Defend the measurement. "The instrument reads. The expert decides. The hashes carry the receipts. If you have a specific case where the diagnostic mis-classifies, I'd like to see it." Then move on.                                                                             |
| <b>Time-running-out handoff.</b>                 | "The projector is open — every country in the corpus, live. The handout has the contact and repo. Thank you."                                                                                                                                                                                                 |

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## Voice and posture reminders

| Setting                             | Action                                                                                                                                      |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Opening</b>                      | No preamble. Walk on, open with "Good morning. <i>Title</i> . The question is X."                                                           |
| <b>Sentence length</b>              | Periods, not commas. Short sentences land.                                                                                                  |
| <b>Numbers</b>                      | Numbers first, qualifier after. "0.21 percent — small" beats "small — about 0.21 percent".                                                  |
| <b>Case studies (6-11)</b>          | Slow there. The cases pay off. On the trajectory slides (7, 9, 11) let the chart breathe — ~40 seconds.                                     |
| <b>Rest-of-world finale (13-18)</b> | A sweep, not six talks. One breath per country. Say the prompt and move. ~20 seconds each.                                                  |
| <b>Close (19)</b>                   | End on " <i>It gives interpretation a reproducible object</i> ," then open the projector and let the live instrument carry the last minute. |
| <b>Hostile question</b>             | Defend the measurement, not the framework. Invite the specific case. Move on.                                                               |

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## Apparatus

- **The deck is one grayscale file** — 19 slides, every slide numbered N / 19. White background, black text, grayscale charts: prints cheaply, reads at a distance.
- **The rest-of-world finale (slides 13-18)** is in the flow, not an appendix — six clean trajectories (deceptive drift present: AUS · CHN · IND; absent: FRA · USA · WLD), one breath each, then the close.
- **Projector** ([codawork2026\\_projector.html](http://codawork2026_projector.html)) — three modes RADAR / BARY / ALIGN + SHOCK overlay. Closes the show; also the Q&A instrument. “Show me a country” → click code → BARY.
- **Full corpus** — the 66-slide, 325-page master PremierDataOutput (all nine countries, all 27 plates each + Stage 2/3 + CNQ) remains in [data\\_outputs/](#) for any question that reaches past the trajectories.
- **Handout** in 6 UN languages — QR on the back leads to all of them.

*The instrument reads. The expert decides. The hashes carry the receipts. The vocabulary holds the line. Five days to Coimbra. Walk to the lectern.*